A Comparative Study: The Impact of AI Chatbot with Retrieval-Augmented Generation On Traditional Customer Service Information Systems in After-Sales Scenario

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Abstract

This study investigates whether AI customer service systems enhanced by Retrieval-Augmented Generation (RAG) and Large Language Models (LLMs) can outperform traditional human service in e-commerce after-sales scenarios. It evaluates response efficiency, problem-solving accuracy, and customer satisfaction using the DeLone and McLean IS Success Model. A quantitative survey is used to compare user experiences and assess AI service effectiveness.

Introduction

Customer service has evolved from human interaction to AI-driven systems. While human agents provide empathy, they are limited in efficiency and scalability. With advances in RAG and LLMs, AI systems can now deliver faster and more accurate support. This study explores whether AI can surpass human service in after-sales contexts.

Research Objectives & Questions

This study aims to compare AI-based and human customer service in e-commerce. It focuses on three aspects:

RQ1: How does AI customer service based on LLM and RAG compare with traditional manual customer service in terms of perceived response efficiency within the after-sales consultation process of e-commerce platforms?

RQ2: How does AI customer service based on LLM and RAG compare with traditional human customer service in terms of perceived accuracy of problem resolution in after-sales service processes?

RQ3: In the after-sales scenario, how do AI chatbots based on LLM and RAG differ from traditional human agent channels in terms of customer satisfaction and perceived service quality?

Hypothesis

H1: AI customer service is perceived as more efficient than human service.
H2: AI customer service is perceived as more accurate in problem resolution.
H3: AI customer service leads to higher customer satisfaction than human service.

Literature Review

Early chatbots improved efficiency but struggled with complex tasks. Recent technologies such as LLMs, RAG, and knowledge integration have enhanced AI capabilities. The DeLone and McLean IS Success Model provides a framework to evaluate system and service performance successfully. However, empirical comparisons between advanced AI and human service remain limited about which Information System are more successful.

Key Terms	*Definitions*
Customer Service Information System	It is a system to design for interaction between the company and customer
Retrieve Augment Generation	A technology that combines external knowledge retrieval with generative artificial intelligence models, by first retrieving relevant information and then generating answers, to enhance the accuracy and reliability of the content.
Large Language Model	It is an artificial intelligence model trained on a large amount of text data, capable of understanding and generating natural language, and used for tasks such as answering questions, writing, and conversation.
DeLone and McLean's information system sucess	A model that assesses the success of an information system by factors such as system quality, information quality and service quality, which affect user usage and satisfaction, and ultimately generate personal and organizational
Artificial Intelligence Customer Service	It is a service to serve customers by AI.
Human Customer Service	It is a service to serve customers by human.

Methodology

This study adopts a quantitative approach using a recall-based survey. Participants evaluate a past after-sales experience with either AI or human service. A between-group design is applied, and data is collected using Likert-scale measures of efficiency, accuracy, and satisfaction to test the hypotheses.